

III. Workforce in the Healthcare Sector in India: Availability and Shortfalls

In this section, we examine the total workforce (medical and non-medical) engaged in the healthcare activities/sector. It consists of the health as well as managerial and financial professionals, along with other skilled workers in the healthcare sector. As mentioned above it is based on the industrial classification (NIC) of workers for the years 1991, 2001 and 2011.

Growth of Workforce in the Healthcare Sector

The total workforce engaged in the healthcare sector in India in 1991 was 1.9 million that rose to 2.35 million in 2001 and further to 4.6 million in 2011 (see Table 1). The addition (increment) of workforce in the healthcare sector during 1991–2001 was merely 0.45 million, but it was the phenomenal addition/increment of 2.25 million that almost doubled the total workforce in the healthcare sector during the decade of 2001–2011. The comparison of the rate of growth in population and that of workforce in the healthcare sector across states indicates that while the rate of growth in population has decelerated during the two decades, between 1991 and 2011, the rate of growth in workforce in the healthcare sector has accelerated (see Table 1). The rate of growth in workforce in the healthcare sector in India was almost similar to that of its population during the 1990s, but it was four times higher than that of the population during 2001–2011.

Health Workers per 1,000 Population: Density as a Measure of Availability and Adequacy

The density when standardized as number of health workers available per 1,000 population, revealed that for three census years (1991, 2001 and 2011) the density was recorded as 2.25 and 2.28 and 3.80 health workers respectively. This trend informs of a marginal improvement on this standardized measure during the 1990s which is quite remarkable during the last decade. This could be due to the rising health workforce, along with a decline in population growth rates during the decade 2001–2011.

Disproportionate Urban Concentration of Health Workers

The density of health workers in rural areas has been lower when compared with that of urban areas. In fact, the density of health workers in urban areas showed a marginal decline during 1991–2001 due to the fact that the rate of growth in urban health workers was not keeping pace with that of urban population during this period. Again, there was a remarkable improvement in terms of growth in rural health workers and its density during 2001–2011. But it could not alter the rural

The ratio of increment in the decade 2001–2011 to the previous decade (1991–2001) is 0.8 in case of absolute increment of population but the increment of health workers in 2001–2011 was nine times higher than that of the increment in the previous decade (1991–2001). In urban areas, such ratio between the last and the previous decade is 1.3 times in case of population and 3.3 times in respect of health workforce.

On the whole, the growth of workforce engaged in the healthcare sector is impressive during the last decade (2001–2011) when compared with the previous decade. But still the rural disadvantage persists in this regard. The urban space is having an explicit relative advantage in this regard where its share of health work force outweighs its share of population which in turn exceeds the minimum need-based requirement. Therefore there is a need for geographical redistribution or redeployment of health workers between rural and urban areas.

Shortage of Health Workers: Reference to WHO's Minimum Requirement

When referred to WHO's standard basic/threshold minimum of 4.45 health workers per 1,000 population, even the density of total workers (including non-medical staff) in the country's healthcare sector falls short of such minimum requirement. Given the size of population in India and WHO's threshold (of 4.45 health workers per 1,000 population), it would have required nearly 3.73, 4.58 and 5.39 million skilled workforce for its healthcare services in India respectively for the years 1991, 2001 and 2011. The shortage of healthcare workers in India is conspicuous; it was 1.85 million in 1991, 2.23 million in 2001 and 0.79 million in 2011 (see Motkuri et al., 2017). As we elaborate in the following sections, the shortfall would be even more if only skilled health professionals, that is, excluding the non-medical staff are considered.

Such a minimum requirement when qualified independently between rural and urban areas, urban areas would far exceed the minimum requirement, leaving the rural space falling far short of it. Visibly such shortage is quite severe in the rural areas when contrasted against the shortage depicted at the national level. Therefore, apart from making up for the shortage in health workforce, there needs to be an emphasis on its redistribution or redeployment.

Surge in Health Workforce during 2001–2011 Explained: ASHAs or Growth of the Private Sector?

The surge in terms of doubling of the workforce engaged in the health sector could be due to growth in the private sector in healthcare services or induction of ASHAs or otherwise. In fact, engaging ASHAs, at least one per village, is one of the important components of the Government of India's National Rural Health Mission (NRHM) launched in 2005. In order to implement NRHM in rural India, a large number of ASHA workers since then have been inducted and made part of

the rural healthcare system. The countryside consists of more than 6 lakh villages and the count would be much more if the number of hamlets is taken into account.

As per the Government of India's report the total number of ASHAs engaged by all the states/union territories (UTs) increased from 7.06 lakh in 2009 to 8.90 lakh in 2013 (GOI, 2014). If we take an approximate of 8.0 lakh ASHAs for the year 2011, they account for little above one-third of the total health workers added (increment) between 2001 and 2011. Their contribution is relatively large in rural areas; it was around two-thirds of increment in rural health workforce during 2001–2011.

The economic reforms introduced in 1991 facilitated growth of the private sector in most of the industries and services sectors including the healthcare services (Duggal et al., 2012). Although the Government of India has since the mid-1980s actively encouraged the formal private healthcare sector through direct and indirect incentives and policy measures, it took shape during the post-reform period (Duggal et al., 2012; see Chakravarthi et al., 2017). In terms of the growth in the private healthcare sector, the 1990s was just a beginning but its impact got multi-fold since the turn of the 21st century. The private healthcare sector in India is a heterogeneous mix ranging from individual practitioners to small and medium hospitals and the corporate commercial hospital sector (Duggal et al., 2012; Chakravarthi et al., 2017). As it is observed, the trend in the healthcare industry during the 2000s indicates that large-scale enterprises in this regard are growing faster than small and medium ones (see Hooda, 2015). For instance, one of the giant corporate healthcare institutions Apollo began its service in the 1980s and expanded multifold geographically within and outside India. Apollo Hospitals' recent annual report claims that it has patients from 120 countries around the globe (see Apollo Hospitals, 2016). The growing demand inducing the healthcare industry and market has led multinational companies to enter into the Indian market and begin their operations (see Chakravarthi et al., 2017).

Therefore, the remarkable growth of workforce engaged in healthcare activities observed for the decade 2001–2011 is partly due to the expanding private healthcare sector in India. Giant corporate bodies are engaging huge number of healthcare professionals and other skilled health workers along with non-medical workers. There has not been much effort in assessing the size and strength of the private healthcare sector especially in respect of its human resources. Apollo Hospitals' recent annual report claims that it has more than 9 thousand beds capacity and nearly 60,000 dedicated healthcare providing staff (including doctors and other paramedics) in its group of hospitals across cities in the country (see Apollo Hospitals, 2016). There are many such corporate and non-corporate multi-specialty hospitals in India, along with clinics and nursing homes. Most of these private healthcare institutions are largely concentrated in urban areas.

It is interesting to note that there is a spatial difference in accounting for these two important factors in surge witnessed in the growth of healthcare workforce in rural and urban areas. While the induction of ASHA workers accounted for the large portion of increment in the rural health workforce during the last decade (2001–2011), it is the growth of the private sector in healthcare that contributed largely for the growth of urban health workforce during this period.

IV. Health Professionals and Skilled Health Workers: NCO-based Workforce

Here we examine the healthcare workforce by differentiating health professionals and other skilled health workers. As mentioned above, it is based on the classification of occupation (NCO) of workers for the year 2001 and 2011 of census data. It needs to be clarified that the total of the NIC-based workforce that engaged in healthcare industry may not exactly match with the total of the NCO-based workforce engaged in this sector. The difference is due to non-medical staff engaged in the healthcare sector being not classified as health workers in the NCO.

Alternative Evidence

In an illuminating in-depth study by Anand and Fan (2016) based on occupational classification (NCO) data of Census 2001, that was carried out at state as well as district level, observed that in India there were 2.1 million health workers, of which 0.82 million (or 39.6%) were doctors, 0.63 million (or 30.5%) were nurses and midwives, and 0.024 million (or 1.2%) were dentists. Of all doctors, 77.2 per cent were allopathic and 22.8 per cent were of AYUSH category. Other categories of health workers were pharmacists, ancillary health professionals, and traditional and faith healers, who comprised 28.8 per cent of the total health workforce. This study observed the density of health workers at the national level was 79.7 doctors per lakh population, 61.3 nurses and midwives per lakh population and dentists were just 2.4 per lakh population (Anand & Fan, 2016). The urban–rural ratio was 1.45 as 59.2 per cent of total health workers were located in urban areas, whereas only 27.8 per cent of the population resided in the rural areas. The study further observed that the ratio of urban density to rural density for doctors was 3.8, for nurses and midwives it was 4.0 and for dentists it was 9.9 (Anand & Fan, 2016).

The same study reveals a startling reality of unqualified nature of most of health workers in the Indian healthcare system. It was observed that of all the cadres of health workers in 2001 less than half (48.6%) of them had secondary education and above. Further, a striking reality was that health workers with medical qualifications (medical diploma/degree) were less than a quarter (23.3%) of the total health workers in India in 2001 (Rao et al., 2016). It is alarming that even among those reported as allopathic doctors about one-third of them had educational level below secondary schooling. Regarding the medical qualification of allopathic doctors, only 42 per cent of them had such medical education. With respect to nurses and midwives, about two-thirds (67%) of them had studied up to secondary level and only 9.9 per cent of them had medical professional qualifications (Rao et al., 2016). It indicates that Indian healthcare system comprises such a large proportion of unqualified workforce.

The study (Anand & Fan, 2016) brought out inter-state differences as follows. There was a six-fold inter-state difference between the highest and lowest density of all health workers (see Anand & Fan, 2016). Certain categories of health

workers were highly concentrated in particular states. For instance, West Bengal had 30.6 per cent of all homeopathic doctors, Uttar Pradesh had 37.5 per cent of all Unani doctors and Maharashtra had 23.0 per cent Ayurvedic doctors. Kerala had 38.4 per cent of all the medically qualified nurses available in the country. In some states the fraction of AYUSH doctors was much higher wherein it was 41.7 per cent in Tripura, 40.5 per cent in Odisha and 38.1 per cent in Kerala (Anand & Fan, 2016). Anand and Fan's (2016) study found a negative correlation⁸ between the percentage of nurses in the health workforce and the percentage of doctors across states. Further, the study found that the density of all health workers in a state was positive but imperfectly correlated with the per capita income of the state⁹. The study infers that better-off states seem to afford more doctors plus nurses per capita¹⁰ and more dentists per capita.¹¹

Further, another study in the recent past by Rao et al. (2016) based on NSSO 68th round EUS in 2011–2012, taking into account occupational classification of workers (NCO), estimated the density of health professionals and skilled health workers. This study considered only those health workers who were qualified while matching the occupation and educational qualification of the workers covered under the survey. This study measured the density per 10,000 population. It is very well discernible from the estimates offered by this study that density of all cadres of qualified health professionals and health workers together observed for the year 2011–2012 appeared to be lower than that observed for the year 2001. In fact, the estimates based on two different sources (Census and NSSO) and methods being strictly not comparable make trend comparison unreliable. The Census 2001 presented combined both the qualified and unqualified health professionals and other health workers whereas the study by Rao et al. (2016) based on NSSO's survey of 2011–2012 presented their estimates of qualified health professionals.

According to the study (Rao et al., 2016) there were 1.4 million unqualified health workers in India representing 56.4 per cent of its total health workforce estimated for the year 2011–2012. The percent of unqualified ones in various cadres of health professionals are as follows: 42.3 per cent of allopathic doctors, 58.4 per cent of nurses and midwives, 27.5 per cent of dentists, 56.1 of AYUSH practitioners and 69.2 per cent health associates (see Rao et al., 2016). Moreover, presence of such unqualified health professionals is quite large in rural areas (71.2%) which surprisingly are to the tune of (48.8%) in urban area as well (Rao et al., 2016). Indeed, prevalence of unqualified and largely unlicensed PMPs attending to certain basic healthcare services is not a rare phenomenon in rural India and there are lakhs of such unqualified doctors in many cases at least one per village (see Narayana, 2004; 2006).

Growth of Skilled Health Workers/Professionals in India during the 2000s

Following the above studies an attempt is made in understanding the changes in the availability of health workers or professionals in India between two Censuses

(2001 and 2011) and that based on NCO. However, due to brevity and limited purpose of present exercise and to avoid repeating the analysis of Anand and Fan (2016), a brief description of the situation is described above. Unlike the Anand and Fan (2016) study where the density of health workers was measured for lakh population, the present study computed health workers per 1,000 population to maintain consistency with the analysis already conducted above based on NIC-based classification of workers. Our focus herein is to present growth and density (per 1,000 population) of NCO based health workers of all cadres together (All) along with that of doctors and nurses including mid-wives (D&NMW), of allopathic doctors (AD), and of nurses including midwives (see Table 3).

What is obvious from the Table 3 is that the size of health workers or various cadres of all the health professionals have increased during the last decade (2001–2011). The total size of skilled health workers that served in India had increased from 2.1 million in 2001 to 3.8 million in 2011 with a net addition of 1.7 million skilled health workers during the period.¹² The rate of growth in all cadres together is 6.3 per cent between 2001 and 2011 (see Table 3). Such a high rate of growth in number of all the skilled health workers in various cadres combined is four times higher than the rate of growth in population (1.64%) during the decade (2001–2011) and hence there was a remarkable improvement in the density of health workers in India during the period. However, some of the sub-categories/cadres of health workers especially AYUSH and pharmacists have registered a negative growth between 2001 and 2011, indicating the decline in the size of health professionals in this cadre/category during the decade. Accordingly, density of these cadres of AYUSH and pharmacists in India has declined.

The negative rate of growth in AYUSH practitioners pulled down the rate of growth in all the doctors including AYUSH and allopathic (to 1.97%). Otherwise, allopathic doctors have registered a rate growth at 3.3 per cent during the period and accordingly their density had improved but it was still less than one per 1,000 population at the national level. When compared to the situation of density of doctors in developed countries such as Germany, UK, and USA and developing countries like China, it is very low in India (see Figure 1). It shows that the density of doctors in India is three to four times lesser than that of developed countries (Motkuri et al., 2017). Indeed the situation of availability of doctors against the requirement in India is abysmally low and should be a matter of grave concern for policymakers in this regard.

There is a very high rate of growth (12.6%) registered for the cadres of nurses and midwives, during the period 2001–2011 (see Table 3). As a result, the share of nurses and midwives in the total skilled health workers in the country had increased from 30.5 per cent in 2001 to 54.3 per cent in 2011. It needs to be mentioned that the cadre of nurses and midwives includes both qualified and unqualified ones. The qualified and trained ones are those that hold appropriate diplomas or degrees in medical sciences that qualify them to engage in the profession (see Appendix). The unqualified are those without any certification (diploma or degree) or training in medical sciences but merely reported as being engaged in this profession. We guess that a major portion of the high rate of growth in nurses and midwives cadre was due to this unqualified nature of professionals in the cadre. In fact, both the studies that were mentioned above